

Evaluation Issues in AI and NLP

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COMP-550

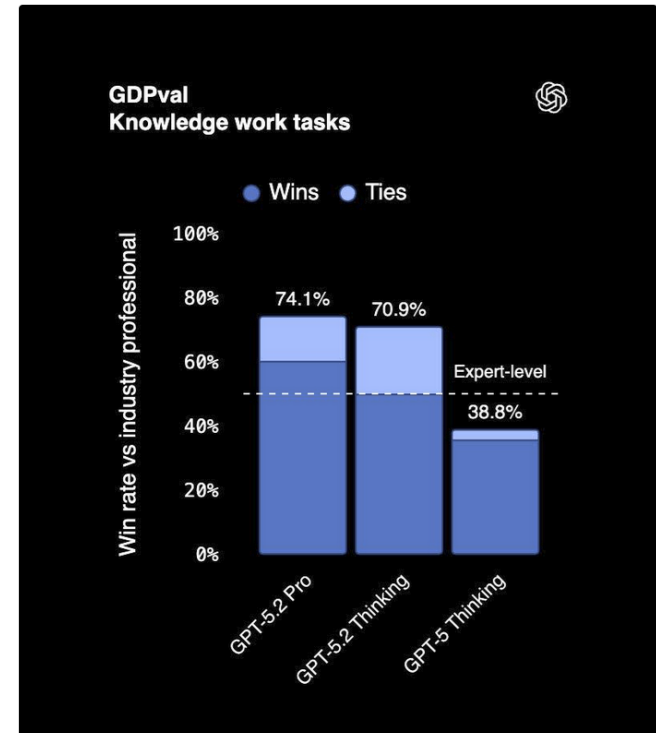
AI Headlines Have Been Driven by Competition

BUSINESS INSIDER

OpenAI says its new GPT 5.2 set a 'new state-of-the-art score' for professional knowledge work

ChatGPT is used by 83% of Moderna employees says Kate Cronin, chief brand officer

Hey, I actually help find the vaccine that we used against



AI Headlines Have Been Driven by Competition

Updated: AUG 27, 2025 9:55 AM ET

The Race for Artificial General Intelligence Poses New Risks to an Unstable World

TECH

AI



ADD TIME ON GOOGLE



The team representing the U.S. has decided to stymie Chinese AI development by blocking all chip exports to China. This has raised the odds, the moderator says, of a Chinese invasion of Taiwan: the U.S. ally that is home to the world's most advanced chip-manufacturing plants. It is 2026, and the simulated world is on the brink of a potentially devastating showdown between two nuclear superpowers.

AI Development Requires Evaluation

Proper evaluations of AI systems necessary to measure progress in the field!

Dominant framing:

- Current benchmarks can measure progress in capabilities
- Competition in order to top these benchmarks

Today: examining some of these assumptions in terms of NLP evaluation!

Evaluation Metrics in NLP

What are some evaluation metrics and methods for different NLP tasks that we have discussed in this class?



Automatic vs. Human Evaluation

What are the advantages and disadvantages of each?

Automatic

- No humans required at test time → often cheaper and fast!
- Could be more consistent; always behave the same way

Human

- Can better “make sense” of the output; they might have better understanding of world in general
- People can be biased → studies cognitive biases
- People can be inconsistent with each other/ themselves; need to calibrate their outputs with respect to each other.

Human Judgments

Absolute or preferences-based?

Text Arena <https://arena.ai/>

Overview

ChatCodeImageVideoStart Voting

View asRankingPareto

Rank byModelsLabs

Categories (27)OverallExpertOccupationalMathInstruction FollowingMulti-TurnCreative WritingCoding

Text ArenaOverall

View overall rankings across various AI models in text-to-text tasks across math, coding, creative writing, and other open-ended domains.

Apr 7, 20265,754,368 votes338 models

Hide Filters

Rank	Rank Spread	Model	Score	Votes	Price \$/M	Context
1	1 - 3	claude-opus-4-6-thinking Anthropic · Proprietary	1503 ±5	15,493	\$5 / \$25	1M
2	1 - 4	claude-opus-4-6 Anthropic · Proprietary	1497 ±5	16,533	\$5 / \$25	1M
3	1 - 6	gemini-3.1-pro-preview Google · Proprietary	1493 ±5	19,482	\$2 / \$12	1M
4	2 - 7	grok-4.20-beta1 xAI · Proprietary	1490 ±7	8,341	N/A	N/A
5	3 - 8	gemini-3-pro Google · Proprietary	1486 ±4	41,615	\$2 / \$12	1M
6	3 - 12	gpt-5.4-high OpenAI · Proprietary	1484 ±7	8,835	\$2.50 / \$15	1.1M
7	4 - 13	grok-4.20-beta-0309-reasoning xAI · Proprietary	1480 ±6	9,003	\$2 / \$6	2M

View all

LLM Evaluation

Emerging area: replace humans with LLMs as judges

- Unreliable at replicating human judgments with large variance in correlations across datasets (Bavaresco et al., 2024).

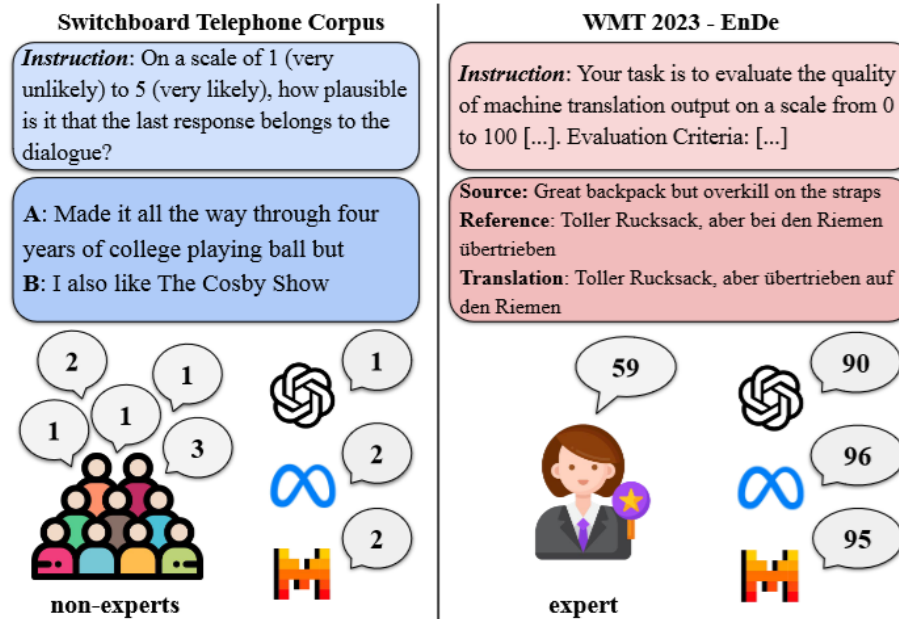


Figure 1: Evaluation by expert and non-expert human annotators and by LLMs for two tasks involving human-generated (left) and machine-generated text (right).

Validity and reliability problems (Chehbouni et al., 2025)

Reference-based vs. Reference-free

Issues to consider:

How are references gathered?

- Expert annotations – costly!
- Crowd annotations – cheaper but need to control quality
- Semi-automatic or LLM-generated labels

References embed assumptions

- Who carries the knowledge or whose knowledge is valued?
- Are there one or many references?

Intrinsic vs. Extrinsic Evaluation

Intrinsic – a model is trained for a task being evaluated w.r.t. the same task.

- Reference-based evaluations are usually intrinsic

Extrinsic – a model trained for a task being evaluated using *another* task

- Evaluate summarization with QA
- Evaluate language modelling using automatic speech recognition

Metric-Driven Research?

Great for stimulating research in an area!

- Standardized test settings (data, metrics)
- Easy to compare systems against each other
- Flashy headlines, build excitement, get funding, ...

Downsides?

- Over-optimising on the target criterion specified by the benchmark (more on this later!)
- Potential to conflate solving a benchmark with solving a task
- If benchmarks are flawed, incorrect decisions based on them lead to harm

Inconsistencies in Model Behaviour

Older results from my lab:

Inconsistent decisions in common sense reasoning

- *Emma did not pass the ball to Janie although she saw that she was open. (she = Emma)*
- *Janie did not pass the ball to Emma although she saw that she was open. (she = Emma still?)*

Trichelair, Emami, Trischler, Suleman, Cheung, EMNLP 2019

Inconsistent "knowledge" about the world:

- *A robin is a [MASK].*
- *Robins are [MASK].*

BERT predicts:

bird

flowers

Ravichander, Hovy, Suleman, Trischler, Cheung, *SEM 2020

Alternative: Evaluation as Measurement

Evaluating an AI system means measuring some property of it

- Capabilities of AI system
- Benefits and harms of AI systems

Learn from other fields that have similar issues!

- Education (e.g., assessing students)
- Psychology (e.g., measuring personality traits)
- Other social sciences (e.g., defining political systems)

Measurement Theory

The scientific study of how to make measurements

- Basic principles and definitions, methods and techniques

Often used in the social sciences, where deciding what and how to measure are difficult research problems.

- E.g., psychology: measurements of personality
- E.g., political science: measurements of characteristics of a political structures
- E.g., education: measurement of student proficiency

Central Concern of M.T.: Validity

Does an evaluation measure what it claims to measure?

Consider this article from the BMJ:

Parachute use to prevent death and major trauma when jumping from aircraft: randomized controlled trial

BMJ 2018 ; 363 doi: <https://doi.org/10.1136/bmj.k5094> (Published 13 December 2018)

Cite this as: *BMJ* 2018;363:k5094

Linked opinion

We jumped from planes without parachutes (and lived to tell the tale)

[Article](#)

[Related content](#)

[Metrics](#)

[Responses](#)

[Peer review](#)

This article has a correction. Please see:

[Parachute use to prevent death and major trauma when jumping from aircraft: randomized controlled trial - December 18, 2018](#)

<https://www.bmj.com/content/363/bmj.k5094>



Fig 2.
Representative study participant jumping from aircraft with an empty backpack. This individual did not incur death or major injury upon impact with the ground

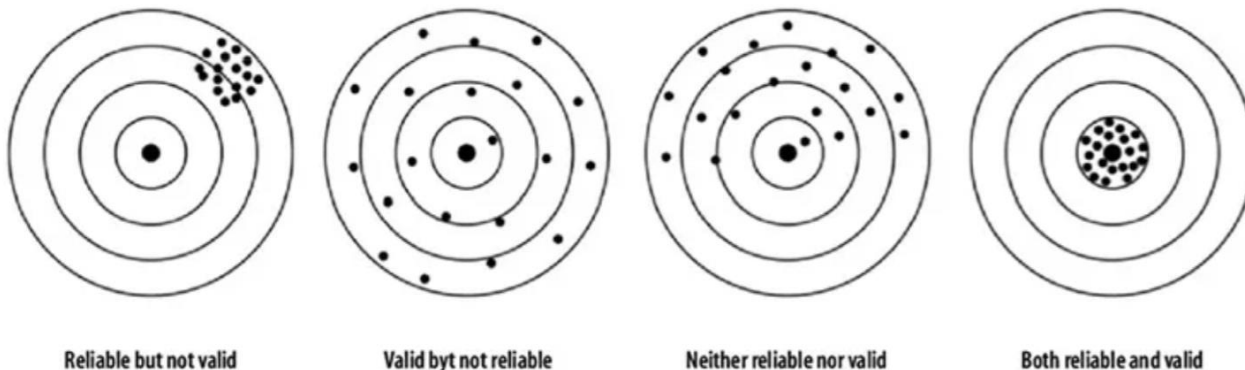
[Download figure](#) [Open in new tab](#) [Download powerpoint](#)

Don't worry, the study
passed ethics review! ;)

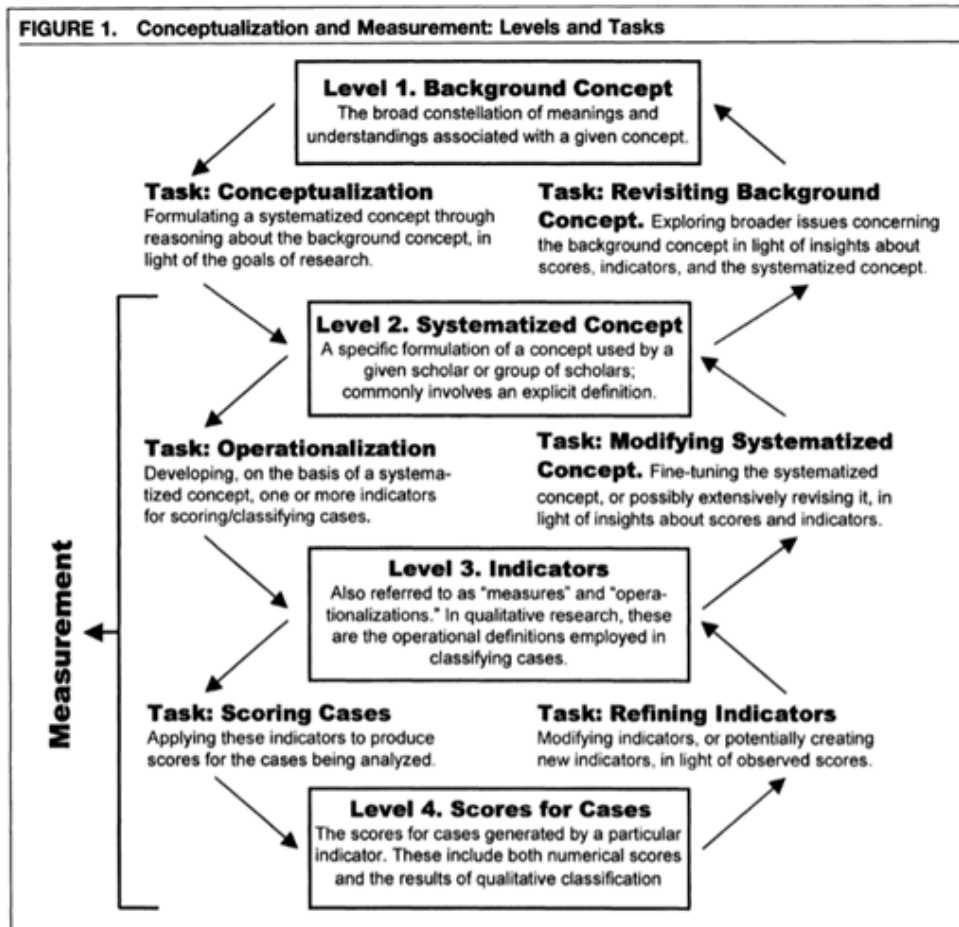
Central Concern of M.T.: Reliability

Can an evaluation's results be repeated under similar conditions?

- i.e., how consistent would the results be?
- Statistical significance is one measure related to reliability
- Other methods/metrics: inter-rater agreement; Cronbach's alpha



Distinguishing Concepts from Measurements



Adcock and Collier,
2001

Evaluating GenAI As Social Science Measurement

Position: Evaluating Generative AI Systems is a Social Science Measurement Challenge

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Abstract

The measurement tasks involved in evaluating generative AI (GenAI) systems are especially difficult, leading to what has been described as “a tangle of sloppy tests [and] apples-to-oranges comparisons” (Roose, 2024). In this position paper, we argue that the ML community would benefit from learning from and drawing on the social sciences when developing and using measurement instruments for evaluating GenAI systems. Specifically, our position is that evaluating

its capabilities (like its mathematical reasoning skills), its behaviors (like regurgitating pieces of its training data), and its impacts (like causing its users to feel harmed). Often, this information takes the form of *measurements* on nominal, ordinal, interval, and ratio scales. Each measurement reflects the amount of some *concept* of interest. Such measurements are obtained via the *process of measurement*, which can involve both qualitative and quantitative approaches.

Across academia, industry, and government (e.g., National Institute for Standards and Technology, 2024; Cooper et al., 2023; Perez et al., 2022; Weidinger et al., 2023), there is an

Wallach et al., 2025, <https://arxiv.org/pdf/2502.00561>

The Turing Test

Alan Turing proposed this as a way to check whether a machine exhibits intelligent behaviour (1950).

- More concrete than some philosophical question such as “Can machines think?”

Setup:

- Human interlocutor chats with an agent, which is either a machine, or another human, for 5 minutes.
- Afterwards, must decide whether the agent is human or machine

Analyzing the Turing Test

What is the construct?

What is the operationalization?

Is this test valid?

Deception in the Turing Test

Most of the effort seems to be to deceive.

Let's consider a transcript from Rose, the 2014 Loebner winner.

Sample Chatlog

Jackie: Where are you right now?

Rose: I'm here, at Bletchley Park.

Jackie: Where's that?

Rose: It doesn't matter.

Jackie: What is the weather there right now?

Rose: It's nice enough out. But it's pretty reasonable the year round.

Jackie: That must be nice. It's pretty cold in Montreal in winter.

Rose: I do prefer nice weather.

Jackie: What do you think about computational linguistics?

Rose: I haven't a clue. San Francisco is perverse. When you expect sunny it's fog and chill winds, but just when you think winter's nearly here, it goes sunny.

Jackie: Do you only care about weather?

Rose: I prefer sunny climates and good weather. Though occasional warm rain is OK.

Goodhart's Law

First formulated in economics:

Once a measure of some quality is turned into a target to optimize, it is no longer a good measure of quality.

Directly optimizing to pass the Turing test leads to systems that “game the test”, reducing the validity of the test in assessing AI progress.

RA4

What are Bender and Koller's claims about the terms "meaning" and "understanding"?

What competing definitions and operationalizations of these terms do you find in the paper?

Recap of Course

What have we done in COMP-550?



Computational Linguistics (CL)

Modelling natural language with computational models and techniques

Domains of natural language

Acoustic signals, phonemes, words, syntax, semantics, ...

Speech vs. text

**Natural language understanding (or comprehension) vs.
natural language generation (or production)**

Computational Linguistics (CL)

Modelling natural language with computational models and techniques

Goals

Language technology applications

Scientific understanding of how language works

Computational Linguistics (CL)

Modelling natural language with computational models and techniques

Methodology and techniques

Gathering data: language resources

Evaluation

Statistical methods and machine learning

Rule-based methods

Current Trends and Challenges

Speculations about the future of NLP



Better Use of *Less* Data

Large amounts of data now available

- Unlabelled, noisy, biased
- May not be directly relevant to your specific problem

We need models and methods that do well even with small amounts of relevant, high-quality data:

- Few-shot learning, in-context learning are partial solutions
- Active learning
- Decomposing task structure

Responsible and Interpretable NLP

Responsible NLP, thinking about benefits and harms of NLP systems

- How do we define these terms?
- Who gets to define these terms?
- How to design proper evaluations?
- Who gets to develop/deploy/regulate/use/be affected by NLP technology?

Interpretable NLP models that can explain their decisions to humans

- The models also need to be able to take feedback to allow people to control how they work (**human oversight**)

Connections to Other Fields

Cognitive science and psycholinguistics

- e.g., model L1 and L2 acquisition; other human behaviour based on computational models

Social sciences; humanities

- Apply NLP techniques to help validate theories
- Soc. sci./humanities to inform NLP (e.g. in evaluation or problem selection)

Human computer interaction and information visualization

- That's nice that you have a tagger/parser/summarizer/ASR system/NLG module. Now, what do you do with it?
- Multi-modal systems and visualizations

Applications of NLP

Healthcare

Finance

Education

Law

Entertainment

Many new potential applications, but they come with risks!

- Anything specific on this slide would be outdated in a few years!

Final Slide

We covered many historical and some current developments in NLP in this course. We hope the intuitions, ideas, and challenges that we presented are useful or at least interesting.

Good luck on your final projects, and have a good summer!